A share selection with fundamentals and machine learning

Historical backtest: 2018-01-02 to 2026-04-29

Does processing improve investable net returns?

The main strategy uses Ridge forecasts and a Top-100 equal-weight portfolio. In 2026 through April 29, Ridge returned 8.8% net, versus 8.2% for the matched benchmark.

Twenty-session forecasts, twenty-session targets and daily portfolio valuation.

The investment idea

Same opportunity set, benchmark and cost treatment

Accounting

Operating strength, growth and financial risk may predict relative returns or subsequent mean reversion.

Technicals

Price trends, volatility and activity may reflect slow adjustment, reversal and liquidity conditions.

Processing

Causal input and score processing may preserve information while reducing unstable rankings and turnover.

A higher fitted IC is not enough. Gross advantage, net excess, risk and turnover must all be evaluated.

The adopted inputs

Definitions and method-specific subsets come from the frozen research specification

Input	Family	Meaning
F1	fundamental	Weighted ROE relative to its 60-day mean
F2	fundamental	ROA trend change, volatility scaled (120 days)
F3	fundamental	Revenue-growth trend change, scaled (120 days)
F5	fundamental	Debt/equity change, volatility scaled (120 days)
T1	technical	20-day momentum
T2	technical	60-day momentum
T3	technical	20-day return volatility
T5	technical	20-day average turnover
F6	fundamental	Signed book-to-market
F7	fundamental	Signed trailing earnings yield
F8	fundamental	Annualized ROA level
F9	fundamental	Operating cash flow to revenue

Union across annual selections; each fold uses mature historical labels. Main methods and raw baselines share the final eligible pool.

Historical data and announcement checks

Data: 2018-01-02 to 2026-04-29 | First actual trade: 2018-07-12

Shanghai and Shenzhen A shares known at the signal date
Exclude then-ST, recent listings and incomplete required inputs
One final shared opportunity set for all main comparisons

401,235 repeated report-slots with future report / announcement dates are masked. This does not prove each value leaked; original unrevised vintages remain uncertified.

The forward close/preclose chain approximates company-action returns. Ordinary cash-dividend arrival dates are not a separate ledger.

The actual adopted processing and model

Ridge | Model and signal construction

Processing: input EMA=0; score EMA=20; neutralization=industry + log market cap; fundamental blend weight=0%

ridge parameters: alpha=10000.0; target=demeaned return; training years=expanding history

Horizon 20 sessions | Targets every 20 sessions | Training stride 5
Annual fitting uses only matured earlier labels; 2018-2020 is fitted history.
Inputs: F1, F2, F3, F5, T1, T2, T3, T5.

Implementation and matched costs

Long only | Top 100 | Equal weight | Next-open execution

Targets refresh every 20 sessions from 2018-07-12
Daily data, signals, valuation and blocked-order retries continue
The schedule never restarts at year-end

The benchmark equally weights the same historical stock pool and uses identical 20-session target dates, trading filters and gross/net cost accounting.

Gross compares with gross, net with net. One-way turnover is (buys + sells) / (2 x opening NAV). Fees remain research assumptions; daily bars do not prove order-queue priority.

Corporate actions create illiquid entitlements

Final sourced ratios | Receivable value is distinct from cash available to trade

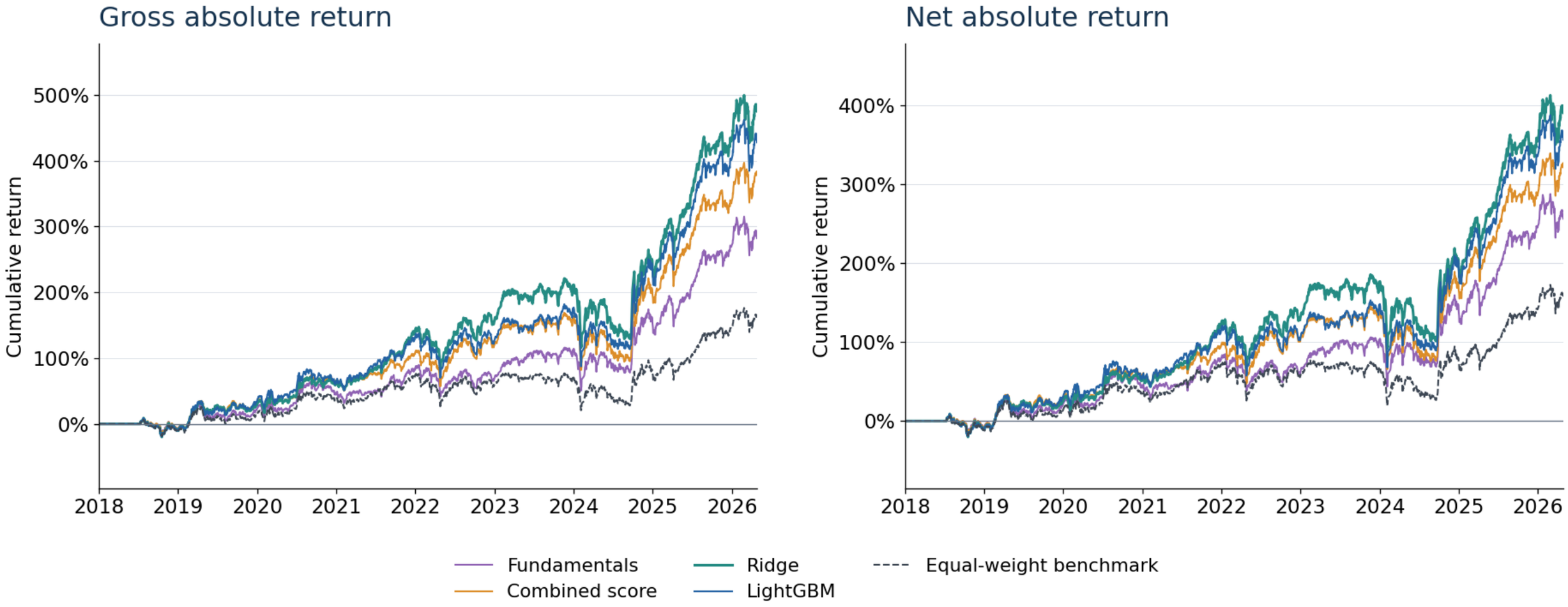
Before successor shares list, the claim keeps the old holding's last market value. It enters NAV but cannot fund buys. Release may create a valuation jump.

All eligible cash-option holdings exercise on the final declaration day. Claims retain the old mark until payment-day close; cash can invest only at the next open.

AVIC / Yulong / Deppon: fifth market session after the completion notice, at day end, is an assumed credit date. Jingwei: S+1 day-end. Unresolved formal delistings use logged zero recovery; this is not proof of actual final recovery.

Historical backtest: gross and net absolute returns

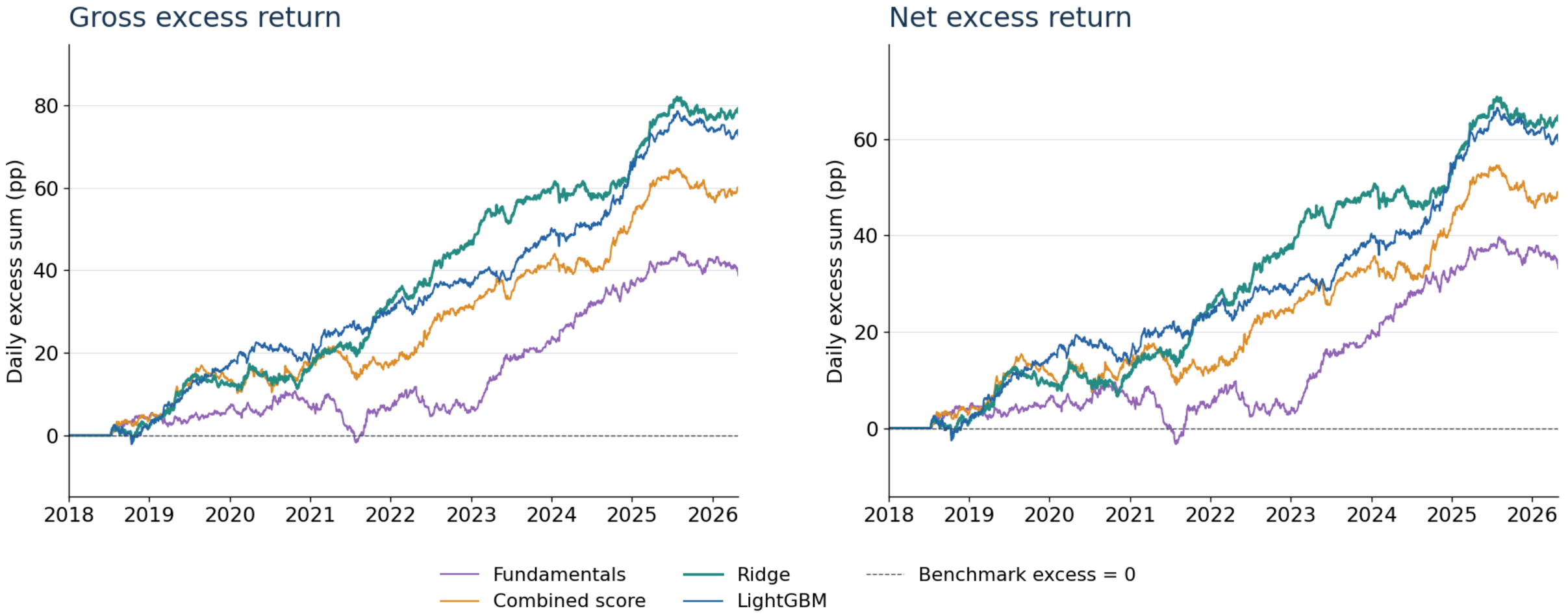
2018-01-02 to 2026-04-29 | 20-session target refresh



Absolute returns compound daily gains and losses. The daily portfolio ledger supplies all plotted values.

Historical backtest: gross and net excess returns

2018-01-02 to 2026-04-29 | 20-session target refresh



Excess sums daily differences in percentage points. The daily portfolio ledger supplies all plotted values.

2021–2025 results

All four methods and the same-cost common-pool benchmark

Method	Gross return	Net return	Net excess pp	Net drawdown
Fundamentals	152.1%	139.9%	+31.9	-30.3%
Combined score	165.9%	145.5%	+33.6	-32.6%
Ridge	226.0%	195.7%	+52.0	-33.0%
LightGBM	204.8%	178.4%	+47.7	-33.7%
Equal-weight benchmark	78.3%	76.6%	+0.0	-32.4%

Ridge daily one-way turnover: 3.4%. The same twenty-session target schedule applies to every method and benchmark.

2026 results to 29 April

All four methods and the same-cost common-pool benchmark

Method	Gross return	Net return	Net excess pp	Net drawdown
Fundamentals	3.8%	3.5%	-4.3	-14.3%
Combined score	10.2%	9.7%	+1.3	-12.3%
Ridge	9.4%	8.8%	+0.6	-12.2%
LightGBM	6.1%	5.6%	-2.4	-13.9%
Equal-weight benchmark	8.2%	8.2%	+0.0	-13.3%

Ridge daily one-way turnover: 3.4%. The same twenty-session target schedule applies to every method and benchmark.

Adopted versus unprocessed: retain the raw comparison

Net cumulative daily excess, in pp | Same final pool and training calendar

Method	2021-25 raw pp	2021-25 adopted pp	2026 raw pp	2026 adopted pp
Fundamentals	+16.1	+31.9	+4.9	-4.3
Combined score	+28.6	+33.6	-3.6	+1.3
Ridge	+23.2	+52.0	-5.2	+0.6
LightGBM	+15.6	+47.7	+0.3	-2.4

Both groups use the same opportunity set, twenty-session target dates and gross/net accounting. Full parameter and return tables are included in the supplementary materials.

Conclusion and limits

Positive 2026 net excess

Main method: Ridge.
2026 net return: 8.8%; daily excess sum: +0.6 pp.
2026 maximum drawdown: -12.2%.

Model overfitting and a short 2026 return record
Report revisions, sample filtering and label availability
Style exposure, trading restrictions and corporate-action marks

Prepared arrays, models, actual research specifications, daily returns and sources support reproduction. Q&A